

Demonstration of Proposed Features

Date	30 june 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation	Predicts the most suitable crop based on soil and environmental parameters.	Implemented	yes	Core functionality working successfully.
2	User Input Form	Accepts Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall values.	Implemented	Yes	User-friendly interface developed.
3	Machine Learning Model Integration	Integrates the trained model.pkl with the Flask backend for real-time prediction.	Implemented	Yes	Model loads and predicts accurately.
4	Prediction Result Display	Displays the recommended crop on the result page.	Implemented	Yes	Results shown correctly after prediction.
5	Input Validation	Validates user inputs and prevents invalid or empty submissions.	Implemented	Yes	Improves application reliability

6	Responsive Web Interface	Provides a clean and responsive interface using HTML and CSS	Implemented	yes	Compatible with desktop and modern browsers.
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Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%